

Multiplication and Division Fact Families

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Name: _____

Date: _____

Learning Intention

I can use multiplication facts for 3, 4, 5 and 10 to write related division facts and explain how the four facts in a fact family are connected.

Strategy Box: same three numbers, four connected facts

Use 4, 6 and 24:

$$4 \times 6 = 24 \quad 6 \times 4 = 24 \quad 24 \div 4 = 6 \quad 24 \div 6 = 4$$

Multiplication builds the total. Division breaks the same total into equal groups.

Worked Example - Read the Array



3 rows of 5 make 15

$$3 \times 5 = 15 \quad 5 \times 3 = 15 \quad 15 \div 3 = 5 \quad 15 \div 5 = 3$$

A. Warm Up - Complete Each Fact Family

1. Numbers: 3, 4, 12 $3 \times 4 = 12$ $4 \times 3 = \underline{\quad}$ $12 \div 3 = \underline{\quad}$ $12 \div 4 = \underline{\quad}$

2. Numbers: 5, 6, 30 $5 \times 6 = 30$ $6 \times 5 = \underline{\quad}$ $30 \div 5 = \underline{\quad}$ $30 \div 6 = \underline{\quad}$

3. Numbers: 4, 7, 28 $4 \times 7 = 28$ $7 \times 4 = \underline{\quad}$ $28 \div 4 = \underline{\quad}$ $28 \div 7 = \underline{\quad}$

Picture Practice - Build Facts from an Array



P1. The array shows ____ rows of ____ = ____ dots.

Write two multiplication facts: _____

Write two division facts: _____

Quick check: In a fact family, the largest number is the total or product.

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B. Build Your Skills - Missing Facts

4. $3 \times 8 = 24$, so $24 \div 3 = \underline{\quad}$ and $24 \div 8 = \underline{\quad}$.5. $5 \times 9 = 45$, so $9 \times 5 = \underline{\quad}$ and $45 \div 5 = \underline{\quad}$.6. $4 \times \underline{\quad} = 36$, so $36 \div 4 = \underline{\quad}$ and $36 \div \underline{\quad} = 4$.7. $\underline{\quad} \times 10 = 70$, so $70 \div 10 = \underline{\quad}$ and $70 \div \underline{\quad} = 10$.

C. Apply It - Solve with Related Facts

8. Four teams have 6 players each. How many players altogether?

Show your strategy:

Multiplication: _____ Related division: _____

9. Thirty-five cupcakes are packed equally into 5 boxes. How many per box?

Show your strategy:

Division: _____ Related multiplication: _____

10. A classroom has 40 chairs arranged in 5 equal rows. How many chairs per row?

Show your strategy:

Division: _____ Related multiplication: _____

11. Explain your thinking

Luca says $32 \div 4 = 6$ because $4 \times 6 = 24$. Explain the error and write the correct fact family.

Multiplication and Division Fact Families - Answer Key

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Learning check

Look for correct use of the same 3 numbers and an explanation that multiplication and division are inverse operations.

A. Warm Up and Picture Practice

1. $4 \times 3 = 12$; $12 \div 3 = 4$; $12 \div 4 = 3$.

2. $6 \times 5 = 30$; $30 \div 5 = 6$; $30 \div 6 = 5$.

3. $7 \times 4 = 28$; $28 \div 4 = 7$; $28 \div 7 = 4$.

P1. 4 rows of 5 = 20; $4 \times 5 = 20$; $5 \times 4 = 20$; $20 \div 4 = 5$; $20 \div 5 = 4$.

B. Build Your Skills

4. 8, 3.

5. 45, 9.

6. 9, 9, 9.

7 7 7 7

C. Apply It

8. $4 \times 6 = 24$ players; related division: $24 \div 4 = 6$ or $24 \div 6 = 4$.

9. $35 \div 5 = 7$ cupcakes; related multiplication: $5 \times 7 = 35$.

10. $40 \div 5 = 8$ chairs; related multiplication: $5 \times 8 = 40$.

11. Sample explanation

Luca used an incorrect related multiplication fact: 4×6 equals 24, not 32.

The correct answer is $32 \div 4 = 8$.

Correct family: $4 \times 8 = 32$; $8 \times 4 = 32$; $32 \div 4 = 8$; $32 \div 8 = 4$.

Teacher feedback checklist

- ☐ Uses the same 3 numbers
- ☐ Writes both multiplication facts
- ☐ Writes both division facts
- ☐ Explains the inverse relationship